

Program Listing Q3a and Q3b shows the code listing of the declaration of Student class and Employee class respectively written in C++ language.

Sem1 2014/15

Q3a(i)

```
1 // in student.h header file
2 #include <string>
3
4 using namespace std;
5 class Student
6 {
7     private:
8         string name;
9         string matric;
10
11     public:
12         Student(string name, string matric);
13         string getMatric() ;
14         void print() ; // print the student name
15 };
```

Program Listing Q3a

```
1 // in employee.h header file
2 #include <string>
3
4 using namespace std;
5
6 class Employee
7 {
8     private:
9         string employer;
10        double wage; // full monthly wage
11
12    public:
13        Employee(string employer, double wage) ;
14        virtual double getWage() ;
15        virtual void print() ; // print the employer name
16 };
```

Program Listing Q3b

Using inheritance, we want to derive a PartTimeStudent class from both Student class and Employee class. This class has an attribute proRate (a double) that indicates the prorated proportion per month. The default prorated proportion per month is 0.5 (half of full monthly wage). The class constructor takes FIVE parameters; employer, wage, name, matric (corresponding to the member attributes of its base classes) and its own member attribute, proRate.

The class also has TWO methods; getWage() and print(). The getWage() function returns the prorated wage as *prorate multiplied by the monthly wage*. The print() function uses the print() functions from its base classes and prints information to the console in the following format :

Employer : <employer name>
Student name: <student name>
ProRate Wage: <result of getWage() >

Write the full declaration and implementation code for PartTimeStudent class. Assume the implementations of both Student class and Employee class are already done.